

ECOMMERCE SALES DASHBOARD

Capstone Project Report

Power BI Dashboard Analysis of E-Commerce Order Data

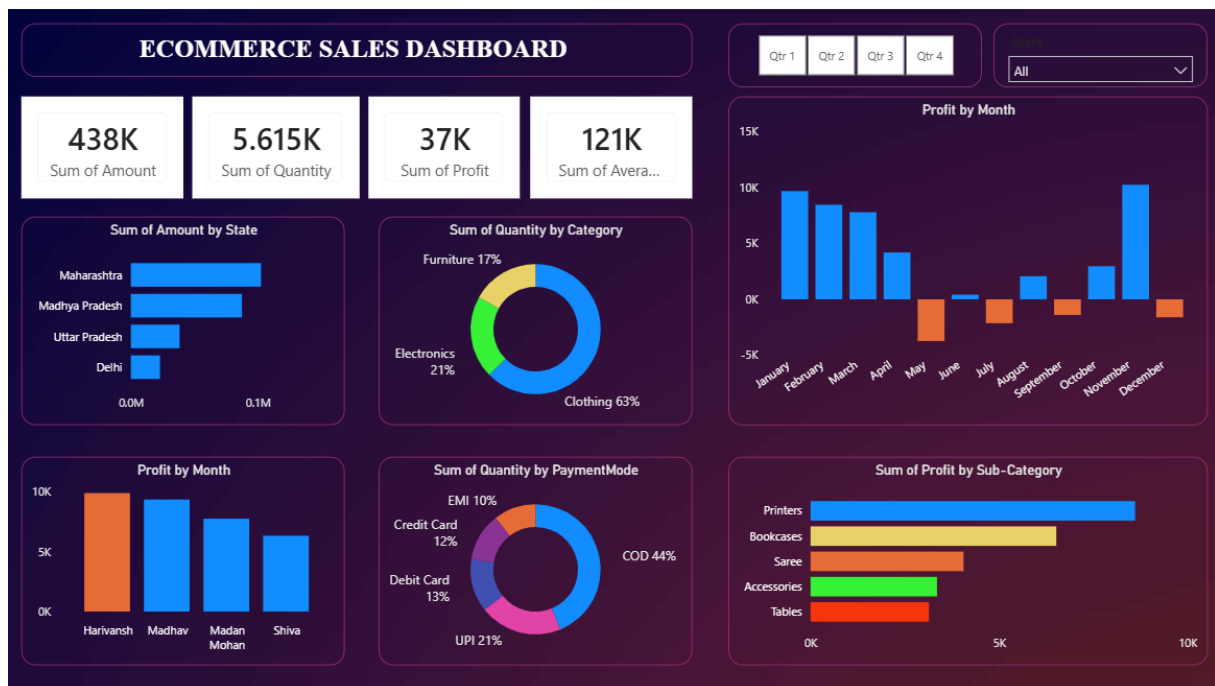


Figure: Ecommerce Sales Dashboard (Power BI)

1. Business Problem

The company generates a large volume of e-commerce order data, including customer orders, transaction amounts, product categories, and payment methods. However, this data was being stored without a centralized, visual way to interpret it. As a result, management had no clear, real-time visibility into overall sales performance, profitability drivers, underperforming products or regions, or customer purchasing behavior.

This lack of visibility made it difficult to take data-driven decisions related to inventory planning, marketing focus, regional strategy, and customer targeting.

Problem Statement

“The company lacked a consolidated, interactive reporting solution to analyze its e-commerce sales performance — making it difficult to identify profitability drivers, underperforming products/regions, and customer trends needed for informed business decisions.”

2. Objectives

- Consolidate order and product-level transaction data into a single, unified data model.
- Track core KPIs — Total Sales, Total Profit, Total Quantity Sold, and Average Order Value — at a glance.
- Analyze profitability across product categories and sub-categories to identify top and underperforming segments.
- Evaluate regional performance by visualizing sales distribution across states.
- Identify top customers contributing the most revenue.
- Understand payment behavior by analyzing order volume across payment modes (COD, UPI, Card, EMI).
- Monitor sales and profit trends over time (by quarter/month) to detect seasonality.
- Enable interactive, self-service exploration through filters and slicers (by date and state) so stakeholders can drill down without technical support.

3. Data Sources

The dashboard was built using two datasets, joined on the common field **Order ID**:

Dataset	Records	Key Fields
Orders.csv	500 orders	Order ID, Order Date, Customer Name, State, City
Details.csv	1,500 line items	Order ID, Amount, Profit, Quantity, Category, Sub-Category, Payment Mode

Data covers order transactions for the year 2018 across multiple Indian states.

4. Key Performance Indicators (KPIs)

Primary KPIs

KPI	Measure	Business Meaning
Total Sales	Sum of Amount	Overall revenue generated from all orders
Total Profit	Sum of Profit	Overall profitability after costs
Total Quantity Sold	Sum of Quantity	Total units/items sold across all orders
Average Order Value	Amount / Orders	Average revenue generated per order

Supporting / Segmented KPIs

- Profit by Month — tracks profitability trend across the year
- Profit by Sub-Category — identifies most and least profitable product lines
- Quantity by Category — shows product mix by volume
- Quantity by Payment Mode — shows customer payment preferences
- Amount by State — shows regional sales distribution
- Profit by Customer — identifies top customers by revenue and profitability

Filters

Order Date (Quarter) and State — allow every KPI above to be recalculated for a specific time period or region.

5. Dashboard Overview

The Power BI dashboard consolidates all KPIs and visuals onto a single page for quick, interactive analysis.

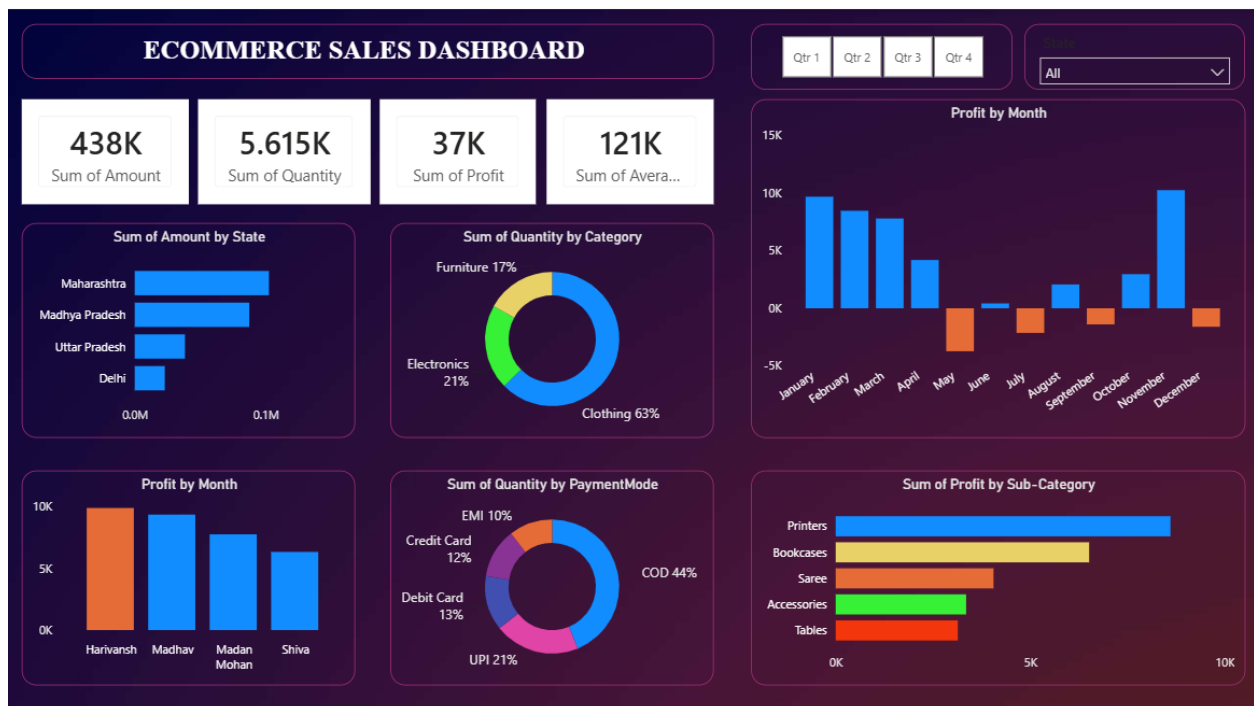


Figure: Ecommerce Sales Dashboard — Power BI

6. Key Business Insights

1. Overall performance is profitable but thin-margin

Total Sales = ₹4,37,771 and Total Profit = ₹36,963, giving an overall profit margin of only about 8.4%. Revenue generation is healthy, but the business retains very little of it as profit.

2. Several sub-categories are losing money

Furnishings (₹806 loss on ₹13,484 sales), Electronic Games (₹644 loss on ₹39,168 sales), Kurti, Skirt, and Leggings all post negative profit. In contrast, Printers, Bookcases, and Saree are the most profitable sub-categories, suggesting pricing or discounting issues concentrated in specific product lines.

3. Furniture is the weakest category overall

Clothing (₹13,325 profit) and Electronics (₹13,162 profit) lead, while Furniture trails at ₹10,476 despite comparable sales, pulled down largely by Furnishings.

4. Cash on Delivery (COD) dominates transactions

COD accounts for 2,456 units and ₹1,55,181 in sales — the largest payment mode by far, nearly double the next closest method (UPI). This creates cash-flow and return-risk exposure.

5. Credit Card is the most profit-efficient payment channel

Credit Card generates ₹86,932 in sales but ₹12,612 profit (about 14.5% margin), nearly matching COD's profit on roughly half the sales volume.

6. Sales are geographically concentrated

Maharashtra (■1,02,498) and Madhya Pradesh (■87,463) together contribute over 43% of total sales, indicating a geographic concentration risk.

7. A high-revenue customer is actually unprofitable

The top customer by sales, Harivansh (■9,902), runs a net loss of –■157, likely from heavy discounting or high-return orders. Vishakha shows a similar pattern.

8. Clear seasonal volatility in profit

Q1 was the strongest quarter (■25,942 profit on ■1,61,288 sales). Several months in Q2–Q3 (May, July, September) posted negative profit despite positive sales, indicating a mid-year slump. November rebounded strongly (■10,253 profit) before December dipped negative again.

7. Recommendations

1. Fix pricing/cost structure for loss-making sub-categories

Audit cost of goods, discount levels, and return rates for Furnishings and Electronic Games. A modest pricing or discount adjustment could turn both profitable with minimal volume loss.

2. Reduce over-reliance on Cash on Delivery

Incentivize prepaid methods such as UPI and Credit Card through small discounts or cashback, since Credit Card already shows a stronger profit-per-sale ratio than COD.

3. Diversify beyond Maharashtra and Madhya Pradesh

Invest in targeted marketing and logistics for the next tier of states (Uttar Pradesh, Delhi, Gujarat) to reduce dependence on two states for nearly half of total sales.

4. Review discounting policy for high-revenue but unprofitable customers

Introduce customer-level profitability tracking, not just revenue tracking, so account-level discounts and offers do not erode margin.

5. Investigate the Q2–Q3 profit dip

Conduct a root-cause review comparing cost, discount, and return data month-over-month to avoid repeating the mid-year loss pattern.

6. Extend Q4 momentum

November was the most profitable month of the year. Plan inventory and marketing spend to extend that festive-season momentum into December, which currently dips back into a loss.

7. Promote high-margin sub-categories

Bundle or cross-sell Printers, Bookcases, and Saree — the most profitable lines — to lift overall margin without needing new customer acquisition.

8. Conclusion

The Ecommerce Sales Dashboard shows a business generating solid revenue (■4.38 lakh) but converting very little of it into profit, with an overall margin of roughly 8.4%. Breaking the numbers down by sub-category, payment mode, geography, customer, and time reveals exactly where that margin is being lost — specific unprofitable product lines, a subset of high-revenue but unprofitable customers, heavy dependence on a low-margin payment method, geographic concentration, and a recurring mid-year seasonal slump.

This is the core value delivered by the dashboard: it shifts the business from asking “are we profitable?” to asking “where exactly are we profitable, and where are we losing margin?” Acting on the recommendations above — correcting loss-making sub-categories, shifting payment mix, diversifying regionally, and addressing the Q2–Q3 dip — can meaningfully improve profitability without necessarily growing total sales.